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Testing a satellite water balance for groundwater recharge in tropical basement catchments

Upper Ouémé, Benin, and Upper Ogun, Nigeria, 2018–2024

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Report version of the web publication at oloyededavid.github.io/Oueme-Ogun-Groundwater-Recharge. Code, data and tests: github.com/oloyededavid/Oueme-Ogun-Groundwater-Recharge.

Summary

Can free satellite data tell us how much rain recharges groundwater where no wells are monitored? In the Upper Ouémé, where the answer can be checked against river flow, the numbers do not add up: 122–169 mm of water a year is unaccounted for.

Rainfall (CHIRPS v2.0, CHIRPS v3.0, TAMSAT v3.1) minus evapotranspiration (FAO WaPOR v3) minus river flow should be close to zero over several years. With river flow estimated from the runoff ratio measured at Bétérou in 1988–2010, it leaves 9.8–13.0% of rainfall in every year of 2018–2024 and with every product.

That water cannot be stored underground; it would need the water table to rise at least 57 m. A higher runoff ratio, of 0.24–0.27, would close the gap, but no recent flow record exists to test it.

The gap is as large as recharge measured in wells in Benin (about 4% and 5–24% of rainfall), so without recent river and well records a satellite balance cannot measure recharge here. The Upper Ogun in Nigeria, with no flow record at all, can be described but not tested.

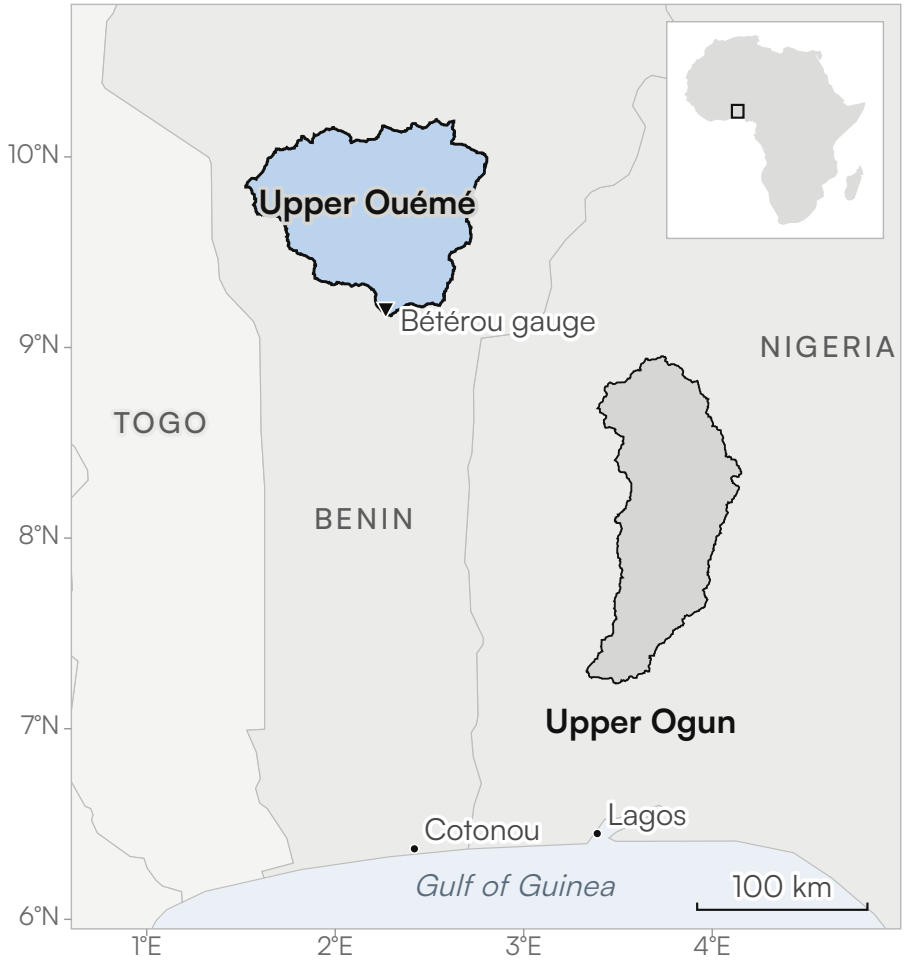
1 Why test it

Rural water in Benin and south-west Nigeria comes largely from weathered crystalline rock, aquifers that across tropical Africa hold little water but are refilled often (MacDonald et al., 2021). Wells in Benin show recharge of about 4% of rainfall (Kotchoni et al., 2019) and 5–24% in the Donga sub-basin of the Upper Ouémé (Kamagaté et al., 2007), but such records are rare. Satellites measure rainfall and evapotranspiration everywhere, so subtracting river flow should leave what recharged the aquifer. Each of those terms, though, has errors of 20–30% (Blatchford et al., 2020; Ogbu et al., 2020), as large as recharge itself. The approach has to be tested where an independent check exists before it is used where none does (Scanlon et al., 2002).

This note asks: does the balance close at Bétérou; if not, what is the leftover water; and what can the same method say about the Upper Ogun?

2 Study area

The **Upper Ouémé** (10,102 km²) drains to the Bétérou gauge in central Benin. It lies on crystalline basement with one rainy season a year (Kodja et al., 2020); groundwater sits in weathered rock above fissured rock, mostly within 100 m of the surface (Lachassagne et al., 2021). The **Upper Ogun** (9,534 km²) is the Ogun basin above the Oyan confluence in Nigeria; no recharge study of it was found. Both were traced from HydroBASINS (Lehner & Grill, 2013).



SELECTED CATCHMENT

Upper Ouémé

Country	Benin
Outlet	River gauge at Bétérou, central Benin
Area	10,102 km²
Study period	2018–2024
Rainfall	1,243–1,298 mm a year
River-flow record	Mean flows 1971–2010 (Kodja et al., 2020)

Select the other catchment on the map, or switch in Figure 2.

Figure 1. The two catchments and the Bétérou river gauge. Select a catchment for its details. Country outlines from Natural Earth.

3 Data and method

Table 1. Data used, as annual totals averaged over each catchment.

Term	Source	Years
Rainfall P	CHIRPS v2.0 and v3.0 (Funk et al., 2015); TAMSAT v3.1 (Maidment et al., 2017)	2018–2024
Evapotranspiration ET	FAO WaPOR v3, 300 m	2018–2024
River flow Q	Mean flow at Bétérou and gauge rainfall (Kodja et al., 2020)	1971–2010

No river flow was measured in 2018–2024, so it is estimated: the published flow divided by gauge rainfall gives a runoff ratio r of 0.142 for 1988–2010, which is applied to each year's rainfall. Whatever is left is the residual ε. It is not called recharge, because it also holds every error in the other terms.

$$Q = r\,P \tag{1}$$

$$\varepsilon = P - ET - Q \tag{2}$$

$$\Delta h = \Sigma\varepsilon \,/\, S_y \tag{3}$$

Rainfall, P · Upper Ouémé, CHIRPS v2.0, 2018–2024 mean: **1,257 mm a year**

Satellite rainfall averaged over the catchment. Three products are used because they disagree.



1,257 − 945 − 178 = **134 mm** left over. This residual holds every error in the three terms; it is not a measure of recharge.

Eq. 3 gives the water-table rise Δh needed to store the residual, for a specific yield S_y. Full method, data inventory and assumptions: research/.

4 Results

4.1 The satellites disagree year to year

The three rainfall products agree on the Upper Ouémé average (1,243–1,298 mm) but not on individual years: CHIRPS v2.0 and TAMSAT do not even agree which years were wetter (r = −0.26 to 0.63 between pairs). In the Upper Ogun they differ by about 222 mm. WaPOR's own rainfall layer is a copy of CHIRPS v3.0, so it adds no information. Both catchments were 4.0% and 8.2% wetter than 1991–2020.

Rainfall

1,243–1,298 mm

Evapotranspiration

945 mm

Rainfall − ET

298–353 mm

Residual ε

122–169 mm

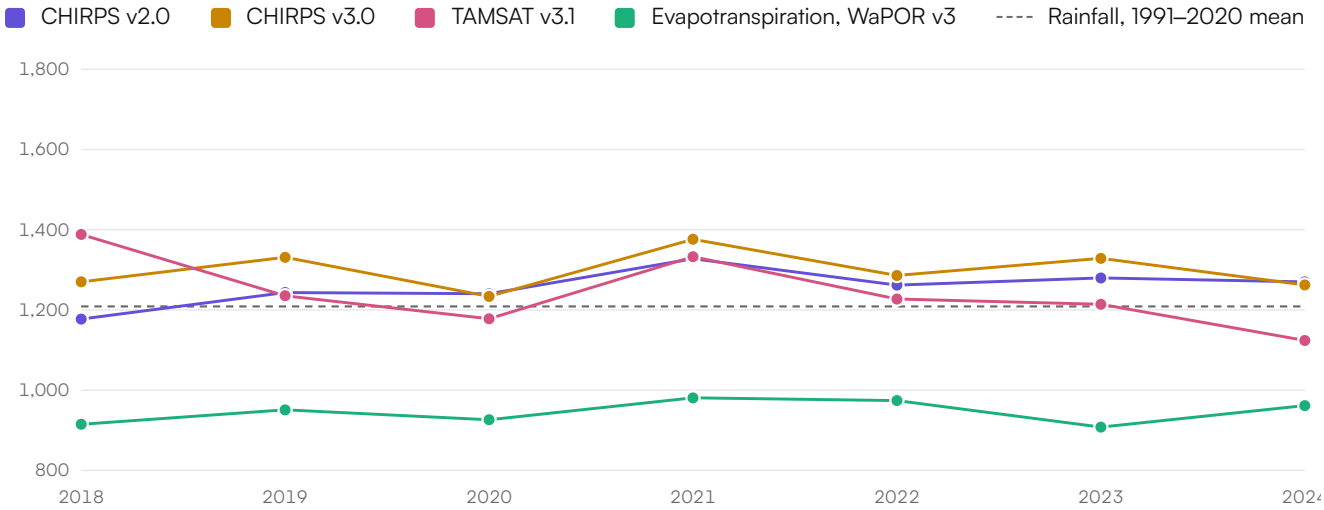


Figure 2. Annual rainfall from three products and evapotranspiration from WaPOR v3, mm per year, for the selected catchment. Dashed: 1991–2020 average rainfall (CHIRPS v2.0).

4.2 The balance does not close

Follow the water through the Upper Ouémé on the map below (Figure 3, Table 2).

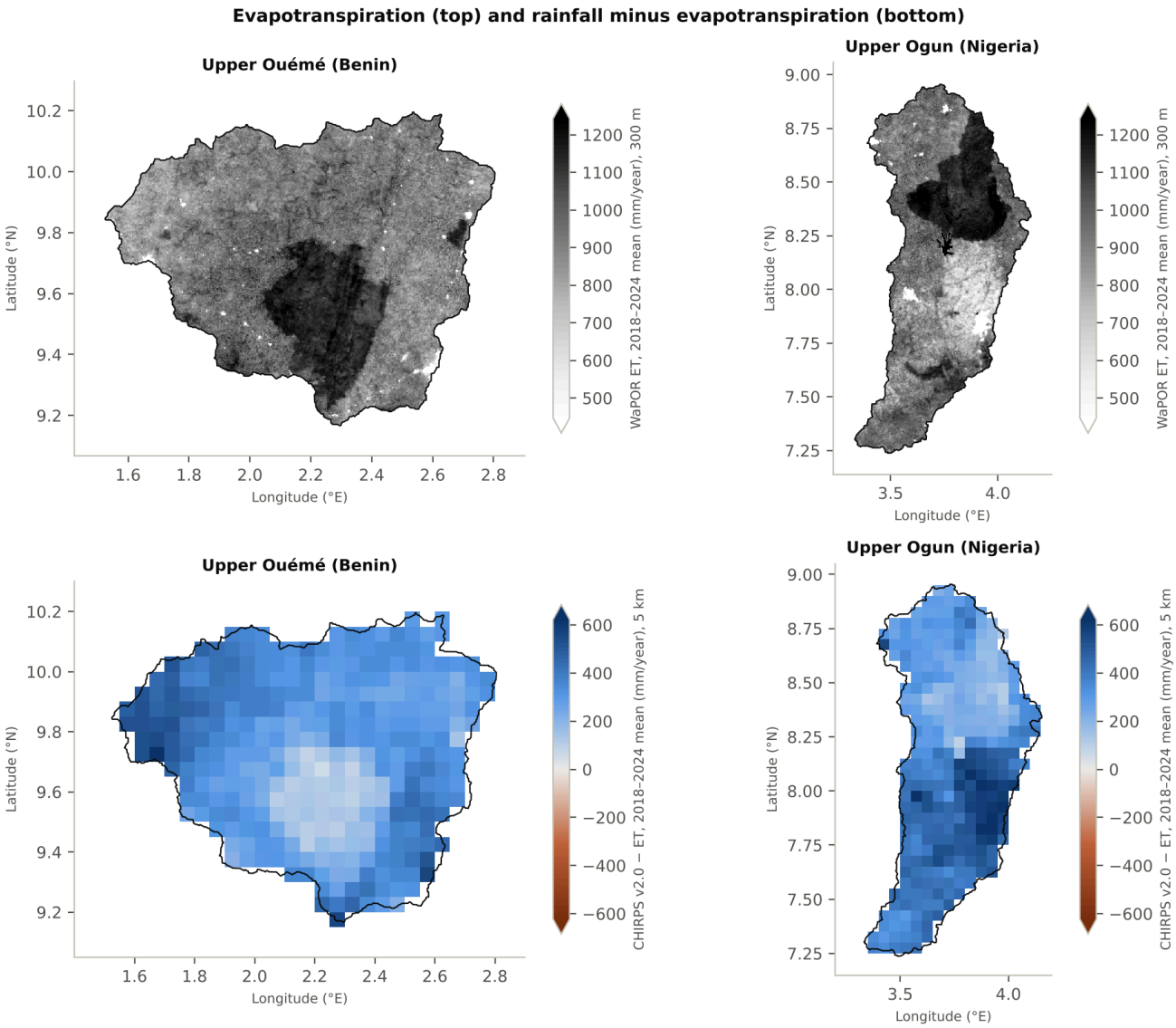


Figure 3. WaPOR v3 evapotranspiration (top) and CHIRPS v2.0 rainfall minus evapotranspiration (bottom), 2018–2024 mean, mm per year, on the 5 km CHIRPS grid. The web version shows these maps year by year.

Table 2. Upper Ouémé balance by rainfall product, 2018–2024 mean, mm per year.

Rainfall product	P	ET	$P - ET$	Q	ε	ε / P, %
CHIRPS v2.0	1,257	945	312	178	134	10.6
CHIRPS v3.0	1,298	945	353	184	169	13.0
TAMSAT v3.1	1,243	945	298	176	122	9.8

4.3 Unless the river carried more

The gap shrinks by about 13 mm for every 0.01 added to the runoff ratio and disappears at 0.24–0.27 (Figure 4). That is 1.7–1.9 times the ratio measured in 1971–2010, but within the 0.14–0.28 seen in single wet and dry years in the Donga sub-basin (Kamagaté et al., 2007). So the river may simply have carried more than assumed. Without recent flow records this cannot be checked.

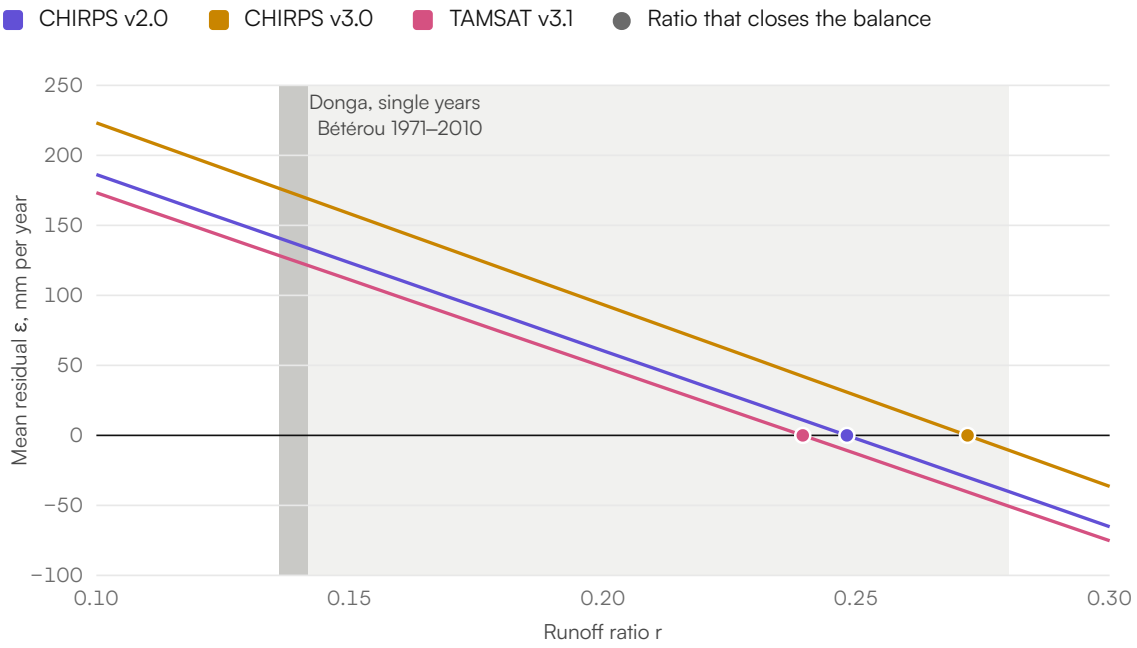


Figure 4. The gap against the runoff ratio assumed. Dark band: ratios measured at Bétérou, 1971–2010. Light band: single years in the Donga sub-basin.

4.4 It is not water stored underground

Over seven years the gap adds up to 937 mm. Stored in weathered rock, with a specific yield of 0.2–1.5% (Lachassagne et al., 2021), that would lift the water table 62–469 m, far above the ground (Figure 5).

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62 m water-table rise needed to store the 2018–2024 residual

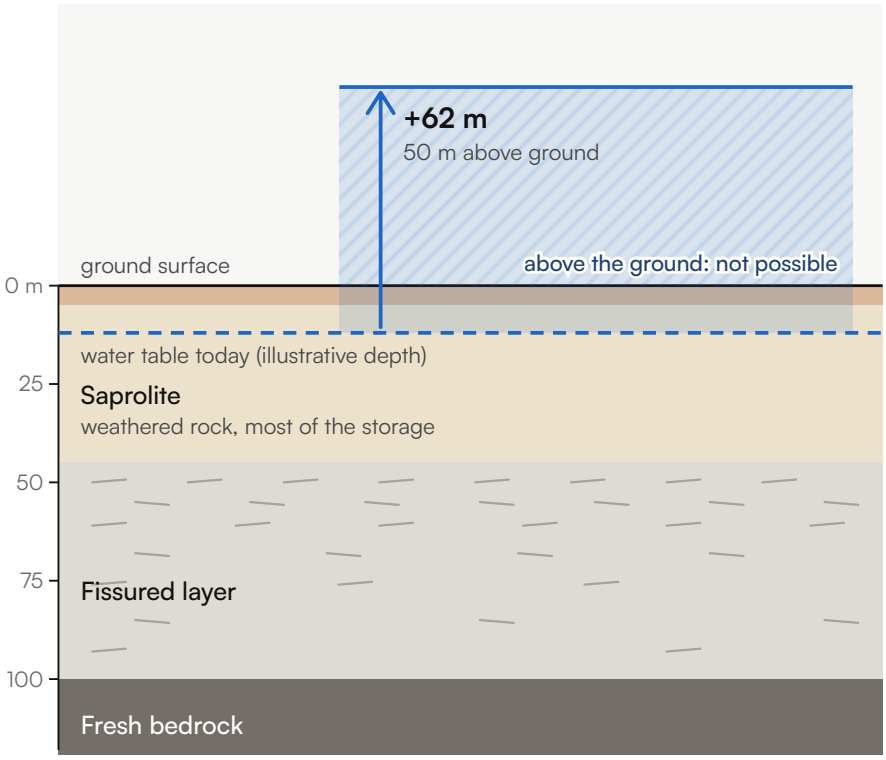


Figure 5. How far the water table would have to rise to hold seven years of the gap (Eq. 3). Schematic profile after Lachassagne et al. (2021); the present water table is drawn at an illustrative depth. Drawn for a specific yield of 1.5% and CHIRPS v2.0; the web version lets both be changed.

4.5 The gap is as large as recharge

At 9.8–13.0% of rainfall, the gap is the same size as recharge measured in Benin's wells (Figure 6). A balance this uncertain cannot measure the recharge it is meant to find.

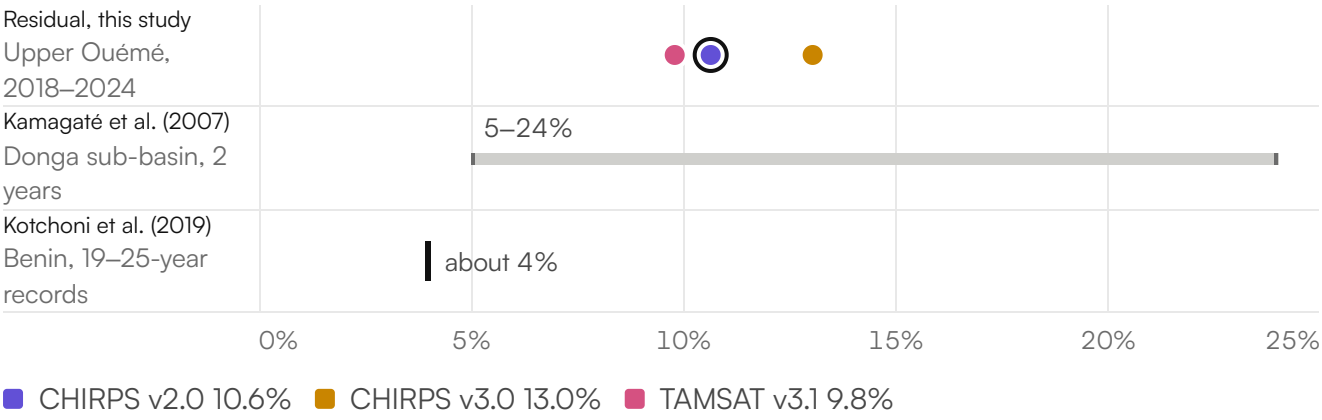


Figure 6. The gap, for each rainfall product, against recharge estimated from wells in Benin, as a share of rainfall.

4.6 Try closing it yourself

Closing the gap needs evapotranspiration 14.2% higher, rainfall 12.4% lower, or a runoff ratio of 0.25 (CHIRPS v2.0). Each is within known error (Blatchford et al., 2020; Ogbu et al., 2020; Kamagaté et al., 2007), so the data cannot say which is wrong.

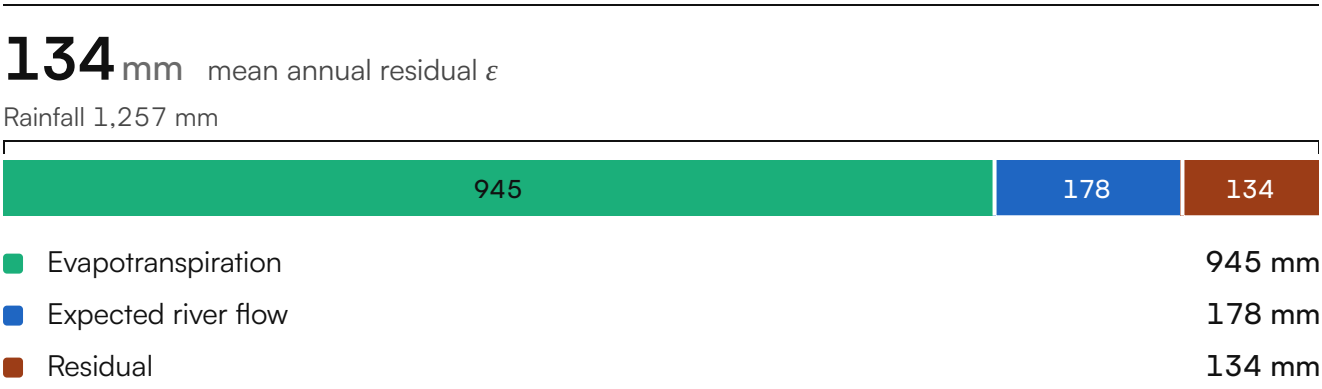


Figure 7. Upper Ouémé water balance, 2018–2024 mean, CHIRPS v2.0, with the runoff ratio of Table 2. The web version lets each term be changed.

5 What it means

The leftover water is not storage, so it is error: most likely river flow that was higher than assumed, or evapotranspiration that WaPOR underestimates. Even a balance that closed would show *net* recharge, because satellite evapotranspiration includes water roots draw back from the water table, while wells show *gross* recharge (Kamagaté et al., 2007). And a gap that is a fixed share of rainfall will look like recharge, which also scales with rainfall (MacDonald et al., 2021), without being recharge. In the Upper Ogun the method gives rainfall minus evapotranspiration of 349–545 mm, untested.

6 Limits

Each row of Table 3 changes one assumption. Seven wet years, rainfall products that share gauges, and an evapotranspiration product not checked on the ground also limit what can be said.

Table 3. How the Upper Ouémé gap changes with each assumption, mm per year.

Assumption	Tested	Gap, mm per year
Rainfall product	CHIRPS v2.0, v3.0 or TAMSAT, r = 0.142 (Table 2)	122–169
Older runoff ratio	1971–1987: r = 0.136	129–177
Ratio from satellite rainfall	Each product's own 1988–2010 rainfall, not gauges	121–179
Runoff ratio of single years	0.14–0.28, Donga sub-basin	−50 to 171
Evapotranspiration	10% higher or lower	27–264
Catchment area	10,070–10,143 km²	121–170

7 Conclusion

With the data available, a satellite water balance does not close at Bétérou, and its error is as large as groundwater recharge. It shows where rain exceeds evapotranspiration, not how much reaches the aquifer. Three records would settle it: river flow at Bétérou for 2018–2024, the AMMA-CATCH well levels in the Upper Ouémé (Galle et al., 2018), for which the analysis code is ready, and ground measurements of evapotranspiration (Mamadou et al., 2016).

Data and code availability

All scripts, the analysis outputs, a dataset inventory (research/data_inventory.csv) and a register of assumptions (research/assumptions.md) are in the repository github.com/oloyededavid/Oueme-Ogun-Groundwater-Recharge. Every number in this report is computed from outputs/summary.json and outputs/annual_balance.csv, which scripts 01–03 produce from openly available data. The Bétérou flows are taken from Kodja et al. (2020); the underlying daily records are not public.

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